

MANAGERIAL ECONOMICS

CLASS: FW27

TERM: FALL25

LECTURES: MONTHLY SATURDAYS 10/11, 11/8, and 12/6 (08:00 AM – 12:00 PM)

LECTURE ROOM: 1E106

PROFESSOR: MARTA SERRA-GARCIA

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OFFICE HOURS:

- Drop-in (office) hours (on Zoom): Wednesdays 1:00PM-2:00PM.
- By appointment (contact me by e-mail and we can set up a meeting)

TEACHING ASSISTANT: Kelvin Leong

TA EMAIL: mleong@ucsd.edu

DESCRIPTION

This course is an introduction to the use of microeconomics in managerial decision-making. We examine how the optimal strategy of a firm is impacted by the structure of the market, the competitive and regulatory constraints, and the structure of information. Topics include supply and demand analysis, price discrimination techniques, contractual relationships, and asymmetric information. Throughout we highlight the importance of **marginal** analysis in decisions.

All the sub-fields of management education (marketing, finance, organizational behavior, operations, information systems, accounting, and strategy) deal in some way with how businesses and other organizations work to allocate scarce resources among competing needs. Because of this, economics can serve as a unifying framework.

OBJECTIVES

The basic objective of this course is to familiarize you with the approach, language, and techniques of managerial economics. You should expect to leave the course with a firm grip on how economists think about business problems, and the ability to apply tools from the course to a variety of problems.

To sum up, the objective of this course is to teach students to ***apply economic principles to solving business problems***.

At the close of Managerial Economics you will be able to:

- Learn marginal analysis and how it can be applied for profit maximization
- Understand consumer demand and how to use price elasticity to guide pricing decisions
- Use versioning, bundling and non-linear pricing as strategies to increase profits
- Understand and compare different types of inputs to production and production costs.
- Derive optimal supply recommendations for firms in competitive markets
- Recognize the effects of regulations on market behavior and welfare
- Apply short-term and long-term strategic thinking for pricing and capacity decisions
- Be cognizant of threats to market entry and how to address them
- Design effective incentive schemes considering conflicts of interest

MATERIALS

Required

- Textbook: *Managerial Economics*, 9th Edition, by William F. Samuelson, Stephen G. Marks, Jay Zagorski.
 - Ebook version (older editions are also fine.)
- Course reader: via Canvas.

COURSE SCHEDULE

Below is an overview of the preliminary course schedule.

Week	Dates	Topic(s)	Reading(s) & Videos	Deliverable(s)*
1	9/29-10/5	Managerial Economics and Marginal Analysis	Week 0 videos Week 1 videos Course reader: Items 1 and 2	Week 1 Reading Questions - 10/6 Week 1 Homework Problems – 10/6
2	10/6-10/12	Demand Analysis and Mark-up Pricing	Week 2 videos Course reader: Items 3 and 4	Week 2 Reading Questions – 10/9
	In person session: 10/11, 8AM-12PM			Week 2 Homework Problems – 10/13
3	10/13 –10/19	Pricing Strategies	Week 3 videos Course reader: Items 5 to 7	Week 3 Reading Questions – 10/20 Week 3 Homework Problems – 10/20
4	10/20 –10/26	Production	Week 4 videos Course reader: Item 8	Week 4 Reading Questions – 10/27 Week 4 Homework Problems – 10/27
5	10/27 –11/2	Costs	Week 5 videos Course reader: Items 9 to 10	Week 5 Reading Questions – 11/3 Week 5 Homework Problems – 11/3
6	11/3 – 11/9	Commodities	Week 6 videos Course reader: Item 11	
	In person session: 11/8, 8AM-12PM			Week 6 Homework Problems – 11/10
7	11/10 – 11/16	Welfare & Market Power	Week 7 videos Course reader: Items 12	Week 7 Reading Questions – 11/17 Week 7 Homework Problems – 11/17
8	11/17 – 11/23	Strategic Thinking	Week 8 videos Course reader: Items 13	Week 8 Reading Questions – 11/24 Week 8 Homework Problems – 11/24
	Online session: 11/18 – 5:30PM	Moblab Competition Simulation	Week 8 video	
9	11/24 – 12/1	Strategy, Deterrence and Collusion	Week 9 videos Course reader: Items 14 to 17	Week 9 Reading Questions – 12/1 Week 9 Homework Problems – 12/1
10	12/2 – 12/7	Information and Risk	Week 10 videos Course reader: Item 18	Week 10 Reading Questions – 12/6
11	In person session: 12/6, 8AM-12PM	Strategy Case, Recap and In-class Final Exam		

IN-PERSON SESSIONS, ONLINE SESSIONS & ASYNCHRONOUS CONTENT

This course will be delivered in two ways:

1. In-person lectures: During these lectures, we will review the materials of the course. I will present key concepts and explain their application. We will discuss cases and news articles. We will have class discussions to share insights and experiences applying economic concepts to business problems.
 - a. The main part of the lectures will be during our Saturday mornings from 8:00AM to 12:00PM.
2. Asynchronous videos: Each lecture is accompanied by a series of short videos in which we discuss key concepts.
 - a. Some of these videos contain material that we do not discuss in class. You are expected to know them and review them at your own pace.
 - b. Other videos contain the same material that we discuss in class. They go deeper into the concept and provide different examples, showing you different ways of solving problems. They can be especially helpful for completing individual homework.
 - c. Some videos will feature guest speakers.
 - d. You are expected to watch all videos.
3. Synchronous online sessions: We will have three synchronous e-sessions, and more can be added as needed for the class.
 - a. Session 1: We will have an initial online meeting to meet each other briefly, discuss the logistics of the class and address initial questions.
 - b. Session 2: We will have an online simulation in which everyone needs to be present, and participate via moblab.
 - c. Session 3: We will have a second online simulation, using moblab.

DELIVERABLES

Reading Questions

- There will be reading assignments for each session. The specific required and suggested readings are listed below. They will be due via Canvas quizzes.
- It is essential that you prepare the readings of Weeks 1, 5 and 9, before our in-person session. During the in-person session we will discuss your answers to the questions asked about the readings and you will be expected to discuss with your fellow classmates. This is a very important part of your learning experience and your participation in class discussion will count towards your class participation grade.
- Reading Questions will be available on Canvas, as Quizzes. The due dates are listed on the table above on several weeks of class.
- These assignments will not be graded by correctness, but for being completed. They will be part of your homework grade.

Homework Problems

Homework assignments are to be handed in individually by each student.

Assignments will be due via Canvas quizzes.

- All assignment Quizzes can be taken **multiple times** – the submission with the highest score counts. Note that your quiz score will be **partial**, based on the questions that can be easily scored, and will miss more complex questions or essay questions.
- The Homework Problems are expected to be a learning tool.

- For some homework questions, with a simple numerical answer, Canvas will let you know if your answers are correct (be aware of formatting issues, that may lead to very minor errors!). This allows you an opportunity to figure out why it was wrong or ask for clarification from us or your classmates, and then re-submit the quiz.
- For other homework questions, you will write a short essay, and our TA will grade it.
- You can submit your calculations, answers to questions, or further work, using the option to UPLOAD a file.
 1. This option is always available at the bottom of the quiz.
 2. To avoid writing answers twice, write “see notes” or “see upload” in the answer field within the quiz, so that we know your answers are uploaded separately when grading.
 3. Please upload only 1 file. You can use different apps to scan your notes. One that may be helpful is HP Smart.

Exam

Final Exam

The Final Exam will be held IN PERSON on Saturday, December 6. The material covered consists of all in-class material, discussions, and homework assignments. The exam is open book, open notes, calculator allowed, and no cell phones. You are expected to understand the message of the readings discussed in class, but not to memorize the facts of the articles or cases.

Class Participation

Class participation includes two components:

- (i) In class (in person) participation. Your in-class participation will be evaluated subjectively and objectively. It will rely upon measures of professional behavior, familiarity with the required readings, and relevance and insight reflected in classroom questions and commentary. Your class participation will be judged by what you add to the class environment, regardless of your technical background. Although several lectures will be didactic, we will rely heavily upon interactive discussion within the class. Students will be expected to be familiar with the required readings. Your participation will be beneficial to you and to your classmates. But be considerate with your classmates’ time – it is the quality of what you say that is most appreciated rather than the quantity.
- (ii) Online participation. We will have moblab simulations, and we will have class discussions on Canvas. Your participation in these will give you additional participation points.

GRADING

Your performance will be evaluated on a final exam, homework assignments, and class participation as follows:

Exams and Deliverables	Percentage
Final Exam	35%
Homework	45%
Class participation	20%
Total	100%

COURSE SUPPORT

Managerial Economics can be a difficult first-course at Rady, as many concepts are new and abstract, and we rely on quantitative and mathematical approaches to solve business problems. We recognize this and have set up many avenues for support for this class.

Work (TA) Sessions

- **Time to be decided.** See Canvas for details on the timing and Zoom room. Work sessions are 1 hour where a TA is be available in person and via Zoom to provide a review of past homework assignments, of the key concepts discussed in class, and practice problems.
- Work sessions are optional, for hands-on, practice and application of concepts, as well as a place where you can ask more detailed questions about specific steps in arriving at different solutions.

Canvas Discussions

- In the course Canvas, there is a Discussions section on the sidebar where students can ask questions and other students can help answer these questions to help and test your learning.
- Both I and a TA will work to provide answers quickly to clarify and answer the questions.

TA Office Hours

- There will be 1 hour of TA office hours each the week. These will be posted on the Canvas homepage.

Professor Office Hours

- I will be available for 1:1 private office hours every week on Wednesdays, and by appointment.

Extensions

- If difficult situations come up for you, please do not hesitate to email me and we can discuss ways to adjust deadlines.

COURSE POLICIES

Taking this class

For this course, we suggest the following:

- Every week watch the videos for the week. They are short 5-15 minute videos that you can listen to and watch while you are exercising, relaxing at home, or doing other activities.
- Every week try to do the homework problems by yourself. If you encounter problems:
 - Refer to the video for that problem (we often list which videos are relevant).
 - Consult with your classmates.
 - Ask on a canvas discussion board, it's likely that others are having the same questions you are.
 - Consult with the TA and with me.
- Every week read the required reading assignments. After you have read them, go straight to the reading questions so that you can answer them easily with the content fresh in your mind.
- You should skim the suggested reading. Your aim in doing this reading before class should be to get a big picture idea of what our class session will focus on. Don't spend time before class trying to understand every little detail in the text. Spend time after class reviewing the parts of the reading that I focused on in class.
- Consult the slides and my notes for any questions or doubts.

On canvas, I also post lecture slides and notes. Prior to class, you will find the lecture slides. After class, I will upload an annotated version of the slides, with all notes that I add during the lecture.

Help

You may need some out-of-class help as the term progresses. There are several places to find it.

First, you can talk to me. Questions right after class, when the questions are fresh in your mind, are often a good idea. Whether it is a specific question, general confusion about a topic or you are just unclear as to where you stand, you can always come and see me in my office.

Next is the course TA. The TA will hold one weekly TA session plus have some office hours. TA sessions are used to review and reinforce material covered in the lectures. TA session attendance is encouraged,

but it is not mandatory. Some students find the TA session period a very efficient time to absorb and reinforce the class material, while other students may prefer to absorb the class material at their own desired time. Office hours are unstructured Q&A sessions - you come with requests to discuss this or that, or (re)work problem X. It may sound like office hours are the better deal, and in extreme cases they are, but many students have found review sessions to be more helpful - hearing classmate's questions and concerns can be enlightening.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/manual/appendices/appendix2.pdf>

ARTIFICIAL INTELLIGENCE / CHATGPT:

Use of AI tools such as ChatGPT, Bing Chat, Bard, Claude, etc. have considerable potential to enhance learning. For instance, Generative AI tools can improve the quality of your writing, identify new sources, provide high-level summaries of complex ideas or terms, and pressure test your arguments for logical inconsistencies. These tools can help you come up with new ideas, overcome writer's block, or refine your original ideas.

You should not use Generative AI tools to generate content that you simply put your name on, solve your case assignments, nor provide answers to your reading questions. This constitutes an academic integrity violation, much like taking another individual's work and misrepresenting it as your own. **If you use Generative AI for an assignment, be transparent about if/how it was used in a footnote or citation** (e.g., link to the ChatGPT chat you created).

- It is the responsibility of each student to ensure that their use of AI tools does not constitute academic misconduct, such as plagiarism or cheating.
- I will assume all students have access to and will use Generative AI and LLMs.
- Generative AI has documented issues with hallucination and inaccuracy (e.g., generating fake citations, failing logical arguments, misrepresenting sources, etc.). You should assume that anything generated by AI is wrong unless it has been fact-checked by you/verified with an independent reliable source.
- You are fully responsible for the content you submit in your assignments and will be graded accordingly.

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or fosorio@ucsd.edu.

Week 1

Key Concepts

1. Value maximization; corporate social responsibility
2. Managerial objectives
3. Fixed and variable cost
4. Revenue and Demand
5. Marginal analysis
 - Marginal benefit
 - Marginal cost
 - Optimization and profit maximization

Learning Objectives

1. Explain what is meant by the value of the firm and indicate the factors that determine it.
2. Defend the assertion that value maximization is a legitimate objective for managers to follow.
3. Define corporate social responsibility.
4. Define and give three business examples of a fixed and a variable cost.
5. Define a firm's revenue and (inverse) demand function.
6. Describe the trade-off between volume and price in demand terms.
7. Explain the concept of marginal revenue and marginal cost.
8. Defend why profits are maximized when marginal profits are zero.
9. Describe how marginal analysis can be a useful tool for managerial decision making.

Videos

1. Week 1 Overview
2. Math Basics - Relevant Concepts
3. Math Basics - Practice Problems
4. Managerial Economics
5. Demand & Revenue
6. Marginal Analysis

Assignments

1. Week 1 Homework Problems
2. Week 1 Reading Assignment

Readings (Required)

1. * "How this Boulder company turned \$2M from crowdfunding into a fully-fledged IoT startup" by Jess Ryan, Built in Colorado, January 9, 2017.
<https://www.builtincolorado.com/2017/01/09/alpine-labs-kickstarter-to-startup>
2. * "Seeking Perfect Prices, CEO Tears Up the Rules," Wall Street Journal, March 27, 2007.
3. Samuelson and Marks, Ch. 2.

Optional Reading: Samuelson and Marks, Ch. 1

* Reading questions in the Reading Assignment

Week 2

Key Concepts

1. Willingness to pay
2. Consumer surplus
3. Self-selection
4. Change in demand vs. quantity demanded
5. Determinants of demand, outside the firm's control: Income and Prices of related goods
6. Elasticity definitions and calculation
7. Price elasticity categories
8. Elasticity and total revenues
 - Relation among P, MR and Ed
 - Optimal markup pricing

Learning Objectives

1. Define willingness to pay and consumer surplus.
2. Calculate the magnitude of consumer surplus in the context of a numerical example.
3. Graphically portray consumer surplus.
4. Explain the concept of self-selection.
5. State the law of demand.
6. Distinguish between a change in demand and a change in quantity demanded.
7. Define and give examples of complements, substitutes and normal goods.
8. Define, precisely, price elasticity of demand.
9. Define cross price and income elasticities.
10. Calculate price elasticity of demand from data.
11. Categorize demand as being elastic or inelastic from given data.
12. Assess the impact on revenues of a price change under alternative scenarios of elasticity.
13. Argue that a firm should not operate in the inelastic range of demand.
14. Correctly provide the equation that relates marginal revenue, price and elasticity.

Videos

1. Week 2 Overview
2. Willingness to Pay and Consumer Surplus
3. Determinants of Demand
4. Relationships Between Products
5. Price Elasticity: Definition
6. Price Elasticity & Revenue
7. Mark-up Rule

Assignments

1. Week 2 Homework Problems
2. Week 2 Reading Assignment

Readings (Required)

1. Samuelson and Marks, Ch. 3 (pages 59-71)
 2. * Drug Goes From \$13.50 a Tablet to \$750, Overnight
 3. * Retailers are charging the same prices across US, boosting income inequality, new research shows.
- * Reading questions in the Reading Assignment

Week 3

Key Concepts

1. Price discrimination
 - Market segmentation
 - Versioning
 - Dynamic price discrimination
 - Bundling
2. Non-linear pricing
 - Two-part pricing

Learning Objectives

1. Describe the objectives and dilemmas of pricing
2. Enumerate and explain the nature of at least three impediments to price discrimination.
3. Explain why versioning accomplishes the same objective as price discrimination.
4. Explain why self-selection is central to successful versioning.
5. Explain what bundling is and name the conditions under which bundling can be profit-maximizing.
6. Describe examples of firms that use dynamic price discrimination.
7. Argue why dynamic price discrimination may be ineffective.
8. Define a two-part pricing scheme.
9. Provide graphically and analytically the optimal two-part pricing scheme from demand and cost information.
10. Explain when two-part pricing is profit-maximizing.

Videos

1. Week 3 Overview
2. Price Discrimination: Definition and types
3. Selection by Indicators
4. Self-Selection
5. Bundling
6. Two-Part Tariffs
7. Case: Fun Fair
8. Total Demand

Assignments

1. Week 3 Homework Problems
2. Week 3 Reading Assignment

Readings (Required)

1. * "Pink Tax: Gender and Other Price Discrimination Factors" Case W04C92, Michigan Ross School of Business
- * Reading questions in the Reading Assignment

Optional Reading:

1. Firm, Prices and Markets, Timothy van Zandt, Ch. 9 & 10
2. Samuelson and Marks, Chapter 3 (71-85)

Week 4

Key Concepts

1. Production function
 - Short run
 - Long run function
2. Law of diminishing returns
3. Marginal product of labor and capital
4. Marginal revenue product of labor
5. Optimal employment of labor in the short run
6. Optimal mix of labor and capital (in the long run)
7. Marginal rate of technical substitution

Learning Objectives

1. Distinguish between short run and long run situations.
2. Give a precise definition and explanation for the principle of diminishing returns.
3. Define marginal product of labor and capital.
4. Employ data on the marginal product of labor and demand for the product (marginal revenue) to compute and explain the concept of marginal revenue product of labor.
5. State and explain what the optimal employment is for a factor such as labor in the short run.
6. Explain why the ratio of the MP's is a measure of the substitutability between factors.
7. Explain verbally, graphically and mathematically why for optimality in the mix of factors, the ratio of the marginal products must equal the ratio of the factor prices.

Videos

1. Week 4 Overview
2. Production: Key Ideas
3. Production Function
4. Production Decisions in the Short-Run
5. Case: Hiring Sales Agents
6. Production Decisions in the Long-Run
7. Least Cost Production

Assignments

1. Week 4 Homework Problems
2. Week 4 Reading Assignment

Readings (Required)

- * "Help yourself", The Economist, July 2, 2009.
- Samuelson and Marks, Ch. 5

* Reading questions in the Reading Assignment

Week 5

Key Concepts

1. Cost types
 - Accounting vs. Economic Cost
 - Opportunity Cost
 - Sunk Cost
2. Cost functions
 - Short run
 - average variable cost
 - average fixed cost
 - marginal cost
 - Long run
 - total cost
 - average cost
3. Shut-down rule
4. Relation between long & short run cost functions: the envelope curve
5. Economies of scale and minimum efficient scale

Learning Objectives

1. Explain the meaning of normal rate of return by relating the concept to opportunity cost.
2. Distinguish among sunk, fixed, and variable costs in a variety of contexts.
3. Distinguish between a short run and a long run situation.
4. Define and derive per unit cost functions (marginal, average variable, average fixed, average total) from knowledge of total cost functions and vice versa.
5. Explain the “shut down” rule.
6. Describe intuitively why the long run average cost curve is the envelope of the short run curves.
7. Define and recognize the existence of economies of scale - given cost data.
8. State the definition of and reasons for both increasing and decreasing returns to scale.
9. Decide which scale of plant is most advantageous for a firm to build.

Videos

1. Week 5 Overview
2. Opportunity Costs
3. Sunk Cost
4. Returns to Scale
5. Shut Down
6. Guest Speaker

Assignments

1. Week 5 Homework Problems
2. Week 5 Reading Assignment

Readings (Required)

Cambridge Software Corp. Harvard Business School Case. (for in-session discussion)

- *As Amazon Pushes Forward With Robots, Workers Find New Roles. New York Times, Sept. 10, 2017
- Samuelson and Marks, Chapter 6

* Reading questions in the Reading Assignment

Week 6

Key Concepts

1. Market structures
 - Price takers
 - Firms with market power
2. Demand facing a price taker
3. Price taker: profit maximizing output
4. Supply curve of a price taker in the short and long run
5. Industry supply curve
6. Relationship between MC and supply
7. Market entry
8. Long run competitive equilibrium
9. Effects of price controls in competitive markets

Learning Objectives

1. Distinguish between a price taker and a price setter, giving two examples of each.
2. Describe and graph the revenue conditions (demand, MR, and TR) facing a price taker.
3. State the profit maximizing condition for the price taker in the short run.
4. Explain how the firm's short run supply curve is determined.
5. Derive the industry supply as the lateral summation of all the firms' supply curves.
6. Predict which firms will stay in the market at the lowest prices.
7. Explain why entry of new firms occurs when price is above average cost for the existing firms.
8. Define, state the conditions, and graphically portray long-run competitive equilibrium for the firm.
9. Explain what is meant by constant, decreasing and increasing cost industries.
10. Distinguish between economies of scale and a decreasing cost industry.
11. Define the long run industry supply price.
12. Predict how the elasticity of supply of inputs will impact the shape of the long run industry supply curve.

Videos

1. Week 6 Overview
2. Competitive Markets: Definition
3. Supply Decisions In Competitive Markets
4. Competitive Markets: Short-Run Equilibrium
5. Competitive Markets: Long-Run Equilibrium
6. Competitive Markets: Moblab Simulation Preparation
7. Alusaf Hillside Case: Introduction
8. Competitive Markets: Market Supply Curves

Assignments

1. Week 6 Homework Problems
2. Week 6 Reading Assignment - No, reading assignment. Reading is part of the Week 6 Homework Problems.

Readings (Required)

- Alusaf Hillside Case
- Samuelson and Marks, Chapter 7

Week 7

Key Concepts

1. Types and characteristics of firms with market power
 - Oligopoly
 - Monopoly
 - Perfect Competition
2. Perfect Competition & Regulation
3. Monopoly & Barriers to entry

Learning Objectives

1. Portray the impacts of price controls, including shortages, surpluses and non-price competition.
2. Distinguish the features of monopoly, oligopoly and monopolistic competition, giving two examples of each.
3. Define the characteristics of price setters
4. Distinguish between the profit-maximizing solution for a monopolist and that for a monopolistically competitive firm.
5. State the conditions for long run monopolistically competitive equilibrium.
6. Give a definition of barrier to entry
7. Enumerate and explain at least six categories of barriers to entry.

Videos

1. Week 7 Overview
2. Alusaf Hillside Case: Resolution
3. Welfare in Perfectly Competitive Markets
4. Welfare in Monopolies
5. Entry Barriers: Natural Monopoly
6. Entry Barriers: Types

Assignments

1. Week 7 Homework Problems
2. Week 7 Reading Assignment

Readings (Required)

1. * "The economic debate over the minimum wage, explained," com, Dylan Matthews, Nov.20, 2019.
2. Samuelson and Marks, Chapter 8

* Reading questions in the Reading Assignment

Week 8

Key Concepts

1. Bertrand (price) competition
2. Entry deterrence
 - Incumbent firm and entrant
 - Entry accommodation vs. predatory pricing
 - Backward induction

Learning Objectives

1. Discuss the incentives of incumbent and entrant firms in the context of entry deterrence
2. Identify the motivation for predatory pricing and the overall manner in which it is achieved
3. Discuss the strategic setting in a situation of market entry with an incumbent
4. Understand the concept of backward induction
5. Explain the pros and cons from predatory pricing vs. accommodation, distinguishing between short term and long term components
6. Bertrand Competition: Show what the equilibrium price and production level is in the Bertrand model.

Videos

1. Week 8 Overview
2. Entry Barriers: Strategic Deterrence
3. Price Competition: Framework
4. Price Competition: Price Wars, Antitrust, and Collusion
5. Quantity Competition: Moblab Simulation Preparation
6. Price Competition: Moblab Simulation Preparation

Assignments

1. Week 8 Homework Problems
2. Week 8 Reading Questions
3. Moblab Synchronous Session (Quantity and Price Competition)

Readings (required)

1. Samuelson and Marks, Chapter 9
2. * “\$1 Pizza Slice Is Back After a Sidewalk Showdown Ends Two Parlors’ Price War” *The New York Times*, Sept. 5, 2012.

* Reading questions in the Reading Assignment

Week 9

Key Concepts

1. Antitrust policies
 - Predatory pricing and abuse of dominant position
 - Chicago School view
 - Collusion
 - Mergers & Acquisitions
2. Oligopoly: Long-term Competition - Cournot competition
 - Strategic thinking
 - Best response production function
 - Equilibrium

Learning Objectives

1. Explain how predatory pricing can be a signal to potential entrants
2. Portray US antitrust policies with regards to abuse of dominant position or monopolization
3. Explain the Chicago School view of entry deterrence
4. Discuss the limitations of the Chicago School view of entry deterrence
5. Understand how collusion may arise
6. Discuss strategies that destabilize cartels and lead to price competition.
7. Explain why mergers are examined by antitrust authorities and how concerns regarding anticompetitive behavior can be addressed
8. Graphically, mathematically and verbally show the profit maximizing quantity of a firm in a Cournot competition market.
9. Describe what a best response production function is.
10. Show analytically what the equilibrium production level and price in a Cournot model is.

Videos

1. Week 9 Overview
2. Deterrence: Antitrust Policy
3. Quantity Competition: Framework

Assignments

1. Week 9 Homework Problems
2. Week 9 Reading Questions

Readings (required)

1. "Airlines and Antitrust: Scrutinizing the American Airlines-US Airways Merger" Harvard Kennedy School, Case 2044.
2. "Justice Department Probes Airlines for Collusion" Wall Street Journal, July 1, 2015.

Optional reading:

1. Modern Industrial Organization, Chapter 19
2. Samuelson and Marks, Chapter 10 (414-422)

Week 10

Key Concepts

1. Asymmetric information
2. Adverse selection
3. Signaling
4. Moral hazard
5. Principal-agent relationship
6. Risk
7. Risk mutualization

Learning Objectives

1. Understand instances in which asymmetric information can undermine the proper functioning of markets.
2. Describe how adverse selection can lead to inefficient market outcomes
3. Understand how signaling can alleviate adverse selection problems.
4. Define moral hazard and provide 3 examples of situations involving moral hazard.
5. Describe mechanisms to alleviate the problem of moral hazard.
6. Explain how moral hazard can undermine insurance markets.
7. Name 3 examples of principal-agent relationships.
8. Understand how certain compensation structures can help alleviate agency problems in employer-employee relationships when worker effort is not perfectly observable by employers.
9. Describe how risk can affect decision making and provide ways in which it can be incorporated into analyses
10. Explain intuitively and analytically how risk mutualization works.

Videos

1. Week 10 Overview
2. Adverse Selection
3. Moral Hazard
4. Risk Sharing
5. Course Review
6. Vivian Garciacano Interview

Assignments

1. Week 10 Reading Questions.

Readings (required)

1. "Bitter Competition: The Holland Sweetener Co vs. NutraSweet". Harvard Business School Case, November 2000.
2. "How did Walmart get cleaner stores and higher sales? It paid its people more." Neil Irwine, The New York Times, 10/15/2016.

Optional

1. Samuelson and Marks, Chapter 14